Contour Integration via Residue Theorem and Cauchy's Integral Formula

Source: J. E. Santos, Cambridge Part IB Complex Methods, Example Sheet 2, Question 9.

1. By integrating the function

$$z^n(z-a)^{-1}(z-a^{-1})^{-1}$$

around the unit circle and applying the residue theorem, evaluate

$$\int_0^{2\pi} \frac{\cos n\theta}{1-2a\cos\theta+a^2}\,d\theta, \tag{1.1}$$

where a is real, $a>1$, and n is a non-negative integer.

2. Obtain the same result using Cauchy's integral formula instead of the residue theorem.

Solution. Parametrise the integration by $z=e^{i\theta}$ with $\theta\in[0,2\pi)$. The integral becomes

$$I=\int_0^{2\pi} d\theta \frac{aie^{in\theta}}{-1+2a\cos\theta-a^2}=\int_0^{2\pi} d\theta \frac{aie^{-in\theta}}{-1+2a\cos\theta-a^2}=-2\pi i\frac{a^{-n+1}}{a^2-1} \tag{1.2}$$

Therefore,

$$\int_0^{2\pi} \frac{\cos n\theta}{1-2a\cos\theta+a^2}\,d\theta=-\frac{I}{ai}=2\pi\frac{a^{-n}}{a^2-1}. \tag{1.3}$$